17 Variational autoencoders

17.1 $\lambda_1, \dots, \lambda_5$ and $\mu_1, \sigma_1^2, \dots, \mu_5, \sigma_5^2$ $\Rightarrow 5 + 10 = 15$

17.2 f concave $\Leftrightarrow \frac{\partial^2 f}{\partial x^2} \leq 0 \;\forall x$ $\leftarrow$ FORMULARY

$$g[x] = \log[x] \Rightarrow \frac{\partial g}{\partial x} = \frac{1}{x}$$

$$\Rightarrow \frac{\partial \frac{1}{x}}{\partial x} = -\frac{1}{x^2} \text{ and } -\frac{1}{x^2} < 0 \;\forall x \setminus \{0\}$$

$$-\frac{1}{x^2} = 0 \text{ for } x = 0 \quad \square$$

17.3 $g[\mathbb{E}[y]] \leq \mathbb{E}[g[y]]$ iff g convex $\leftarrow$ FORMULARY

g convex $\Leftrightarrow \frac{\partial^2 g}{\partial x^2} \geq 0 \;\forall x$

$$g[x] = x^{2n} \Rightarrow \frac{\partial g}{\partial x} = 2n\, x^{2n-1}$$

$$\Rightarrow \frac{\partial^2 g}{\partial x^2} = 2n(2n-1)\, x^{2n-2}$$

$$= 4n^2 x^{2n-2} - 2n\, x^{2n-2} = (4n^2 - 2n)\, x^{2n-2}$$

proof by induction: $n = 1$ $(4-2)x^0 = 2 \geq 0$

$n \to n+1$: $(4(n+1)^2 - 2(n+1))\, x^{2(n+1)-2}$

$$= (4n^2 + 4n + 4 - 2n - 2)\, x^{2n}$$

$$= (4n^2 + 2n + 2)\, x^{2n} \geq 0$$

As x^{2n} positive $\forall n \in \{1, 2, \dots\}$ (even number), this holds!

$\Rightarrow \mathbb{E}[x]^2 \leq \mathbb{E}[x^2]$.

Set $n = 1$. Then $g: x \to x^2$ and the above follows immediately $\square$

17.4
$$\mathrm{ELBO}[\theta, \phi] = \int q[z|\theta] \log\left[\frac{Pr[x, z|\phi]}{q[z|\phi]}\right] dz$$

$$= \int q[z|\theta] \log(Pr[x|z, \phi]) dz - D_{KL}[q[z|\theta] || Pr[z]] \quad (17.18)$$

$$D_{KL}[q[z|x] || Pr[z|x, \phi]] = \int q[z|x] \log\left[\frac{q[z|x]}{Pr[z|x, \phi]}\right] dz$$

$$\overset{17.19}{=} \int q[z|x] \log\left[q[z|x]\left(\frac{Pr[x|z, \phi]\, Pr[z]}{Pr[x|\phi]}\right)^{-1}\right] dz$$

$$= \int q[z|x] \log\left(q[z|x] \left(Pr[x|z, \phi] Pr[z]\right)^{-1} Pr[x|\phi]\right) dz$$

$$= \underbrace{\int q[z|x] \log\left[\frac{q[z|x]}{Pr[z]}\right] dz}_{D_{KL}[q[z|x] || Pr[z]]} + \overbrace{\int q[z|x] \log[Pr[x|\phi]] dz}^{\int q[z|x] dz = 1} \underbrace{- \int q[z|x] \log[Pr[x|z, \phi]] dz}_{= \mathbb{E}_{q[z|x]}[\log[Pr[x|z, \phi]]]}$$

$$\Leftrightarrow \mathbb{E}_{q[z|x]}[\log(Pr[x|z, \phi])] - D_{KL}[q[z|x] || Pr[z]] \leq \log[Pr[x|\phi]]$$

$$= \log[Pr[x|\phi]] - D_{KL}[q[z|x] || Pr[z|x, \phi]] \quad \square$$

17.5

$$\frac{\partial}{\partial \phi} \mathbb{E}_{Pr[x|\phi]}[f[x]] = \frac{\partial}{\partial \phi} \sum_x f[x]\, Pr[x|\phi] \quad (x \text{ discrete})$$

$$= \sum_x f[x] \underbrace{\frac{\partial}{\partial \phi} Pr[x|\phi]}_{= Pr[x|\phi] \frac{\partial}{\partial \phi} \log[Pr[x|\phi]]}$$

FORMULARY $\frac{\partial}{\partial \phi} \log Pr = \frac{1}{Pr} \frac{\partial Pr}{\partial \phi}$

$$= \sum_x f[x] \frac{\partial}{\partial \phi} \log[Pr[x|\phi]]\, Pr[x|\phi]$$

$$= \mathbb{E}_{Pr[x|\phi]}\left[f[x] \frac{\partial}{\partial \phi} \log[Pr[x|\phi]]\right]$$

FORMULARY: MC approx.!

Using Monte Carlo approximation: $\mathbb{E}_{Pr[x|\phi]}[g[x]] \approx \frac{1}{I} \sum_{i=1}^{I} g[x_i]$

with $g := f[x] \frac{\partial}{\partial \phi} \log[Pr[x|\phi]]$

we arrive immediately at: $\frac{\partial}{\partial \phi} \mathbb{E}_{Pr[x|\phi]}[f[x]]$

$$\approx \frac{1}{I} \sum_{i=1}^{I} f[x_i] \frac{\partial}{\partial \phi} \log[Pr[x_i|\phi]] \quad \square$$

17.6 It is better to use spherical linear interpolation rather than regular linear interpolation when moving between points in the latent space because of the curse of dimensionality. (everything is far apart in higher dimensions!)

17.7 $Pr[x|\theta] = \sum_{k=1}^{K} \lambda_k \mathrm{Norm}_x[\mu_k, \Sigma_k]$

$$Pr[x|z, \theta] = \mathrm{Norm}_x[\mu_z, \Sigma_z]$$

and $Pr[z|\theta] = \mathrm{Cat}_z[\lambda]$

so
$$Pr[x|\theta] = \sum_{k=1}^{K} Pr[x, z = k|\theta]$$

$$= \sum_{k=1}^{K} Pr[x|z = k, \theta] \cdot Pr[z = k|\theta]$$

$$= \sum_{k=1}^{K} \lambda_k \mathrm{Norm}_x[\mu_k, \Sigma_k]$$

Update rules:
$$\lambda_k^{[t+1]} = \frac{\sum_{i=1}^{I} r_{ik}}{\sum_{j=1}^{K} \sum_{i=1}^{I} r_{ij}}$$

$$\mu_k^{[t+1]} = \frac{\sum_{i=1}^{I} r_{ik} x_i}{\sum_{i=1}^{I} r_{ik}}$$

$$\Sigma_k^{[t+1]} = \frac{\sum_{i=1}^{I} r_{ik}(x_i - \mu_k^{[t+1]})(x_i - \mu_k^{[t+1]})^T}{\sum_{i=1}^{I} r_{ik}}$$